

What are Joins in SQL?

Joins in SQL are used to combine rows from two or more tables based on a related column between them. Joins help retrieve data that is spread across multiple tables.

Types of Joins:

INNER JOIN:

Returns only the rows that have matching values in both tables.

For Example, we have **Customers** table

CustomerID	Name
1	Ali
2	Sana
3	Usman

And Second is **Orders** Table

OrderID	CustomerID	Product
101	1	Laptop
102	2	Phone
103	4	Monitor

Query

```
SELECT Customers.Name, Orders.Product  
FROM Customers  
INNER JOIN Orders
```

```
ON Customers.CustomerID = Orders.CustomerID;
```

Name	Product
Ali	Laptop
Sana	Phone

What is LEFT JOIN in SQL?

LEFT JOIN is used to combine rows from two tables, and it:

- Returns all rows from the left table
- Returns matching rows from the right table
- If there is no match, it shows NULL for the right table's columns

For Example, this students table is Left Table

StudentID	Name
1	Ali
2	Sana
3	Usman

And Marks is the Right Table

StudentID	Subject	Marks
1	Math	85
2	Math	90

LEFT JOIN Query:

```
SELECT Students.Name, Marks.Marks
```

```
FROM Students
```

```
LEFT JOIN Marks
```

```
ON Students.StudentID = Marks.StudentID;
```

Name	Marks
Ali	85
Sana	90
Usman	NULL



What is RIGHT JOIN in SQL?

RIGHT JOIN is used to combine rows from two tables, and it:

- Returns all rows from the right table
- Returns matching rows from the left table
- If there is no match, it shows NULL for the left table's columns

For Example, Customers (Left Table)

CustomerID	Name
1	Ali
2	Sana
3	Usman

And Orders (Right Table)

OrderID	CustomerID	Product
101	1	Laptop
102	2	Phone
103	4	Monitor

Query

```
SELECT Customers.Name, Orders.Product
```

```
FROM Customers
```

```
RIGHT JOIN Orders
```

```
ON Customers.CustomerID = Orders.CustomerID;
```

Result of the Query

Name	Product
Ali	Laptop
Sana	Phone
NULL	Monitor